Differential Geometry Excercise 1

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Question 1

Let M and N be smooth manifolds. For every chart $\varphi_i : U_i \to R^m$ where $U_i \subseteq M$ and chart $\varphi_j : U_j \to R^n$ where $U_j \subseteq N$, define a chart

$$\varphi_{i,j} : U_i \times U_j \to R^{m+n} \qquad \varphi_{i,j}(p,q) = (\varphi_i(p), \varphi_j(q)) \tag{1}$$

where $(\cdot,\cdot)$ denotes coordinates concatenation.
Now let $\{U_i, \varphi_i\}$ and $\{V_j, \psi_j\}$ be atlases in the respective structures of M and N. Then the set

$$A = \left\{\left(U_i \times V_j, \varphi_{i,j}\right)\right\} \tag{2}$$

covers every point on $M \times N$. In addition, every chart in A is smooth. So, for every two subsets of $M \times N$ that coincide, $(U_i \times V_j) \cap (U_k \times V_l) \neq \emptyset$, the transition map from one subset to the other

$$\varphi_{k,l} \circ \varphi_{i,j}^{-1} = (\varphi_k \circ \varphi_i^{-1}, \varphi_l \circ \varphi_j^{-1}) \tag{3}$$

is also smooth. Thus, A is an atlas and it generates a smooth structure over $M \times N$, making it a smooth manifold with dimension $\dim(M) + \dim(N)$.

Question 2

Let $f : M \to N$ be an immersion that is also a submersion. Because it is an immersion, for every point $p \in M$ its differential Df_p is injective. Because it is an submersion, for every point $p \in M$ its differential Df_p is surjective. Thus, Df_p is invertible for every point $p \in M$.
According to the Inverse Function Theorem, for a smooth function $f : U \to \mathbb{R}^n$ where $U \subset \mathbb{R}^n$ for which Df_p is invertible at point $p \in U$, there exist neighborhoods V, W of p and $f(p)$ respectively such $f : V \to W$ is a local diffeomorphism.

Question 3

Let $f = (f_1, ..., f_k) : M \to \mathbb{R}^k$ be a submersion. Therefore its differential $Df_p : T_pM \to \mathbb{R}^k$ at each point p is surjective, and spans $\mathbb{R}^k$.
The differential in terms of coordinates $\left\{(df_i)_p : T_pM \to \mathbb{R}, i = 1, ..., k\right\}$ are k linear maps. Let us assume that they are linearly dependent. This means that there exist a constant vector $u \in \mathbb{R}^k, u \neq 0$ such that

$$u \cdot (df_i)_p = 0 \tag{4}$$

So, for every vector $v \in T_pM$

$$u \cdot (df_i)_p (v) = 0 \tag{5}$$

but since Df_p is surjective

$$\left\{(df_i)_p\,(v) : v \in T_pM\right\} = \mathbb{R}^k \tag{6}$$

and so, for every vector $w \in \mathbb{R}^k$, $u \cdot w = 0$, which means $u = 0$. Therefore, the only linear combination of $(df_i)_p$ that vanished identically is the trivial one, and they are linearly independent.

Question 4

$\mathbb{R}P^n$ is the set of all lines passing through the origin in $\mathbb{R}^{n+1}$:

$$\mathbb{R}P^n = \left\{\{\lambda x : \lambda \neq 0\} : x \in \mathbb{R}^{n+1}\backslash\{0\}\right\} \tag{7}$$

Then, let $U_i = \{\{\lambda x : \lambda \neq 0\} : x \in \mathbb{R}^{n+1}\backslash\{0\}, x_i \neq 0\}$. For each such open set U_i, define a bijection $\varphi_i : U_i \to \mathbb{R}^n$ such that

$$\varphi_i(\{\lambda x : \lambda \neq 0\}) = x_i^{-1}(x_1, ..., x_{i-1}, x_{i+1}, ..., x_{n+1}) \tag{8}$$

The inverse of this bijection is clear:

$$\varphi_i^{-1}(x) = \{\lambda y : \lambda \neq 0\}, \qquad y = (x_1, ..., x_{i-1}, 1, x_{i+1}, ..., x_{n+1}) \tag{9}$$

The set of all charts $\{\varphi_i : i = 1, ..., n+1\}$ covers $\mathbb{R}P^n$. The image of each bijection $\varphi_i(U_i)$ is an open set, and φ_i is a homeomorphism, therefore $\varphi_i(U_i \cap U_j) \quad \forall \quad i, j = 1, ..., n+1$ is also an open set. Finally, for each $i, j = 1, ..., n+1$:

$$\varphi_i \circ \varphi_j^{-1}(x) = \begin{cases} x_i^{-1}(x_1, ..., x_{i-1}, x_{i+1}, ..., x_{j-1}, 1, x_{j+1}, ..., x_{n+1}) & i < j \\ x_i^{-1}(x_1, ..., x_{i-1}, x_{i+1}, ..., x_{n+1}) & i = j \\ x_i^{-1}(x_1, ..., x_{j-1}, 1, x_{j+1}, ..., x_{i-1}, x_{i+1}, ..., x_{n+1}) & j < i \end{cases} \tag{10}$$

Which is C^∞ since $x_i \neq 0$. Thus, $\{\varphi_i\}$ is an atlas, and the smooth structure it defines makes $\mathbb{R}P^n$ a smooth manifold.

Question 5

Let $\pi : S^n \to \mathbb{R}P^n$ be the map that associates to a point $x \in S^n$ the line through the origin that passes through x:

$$\pi(x) = \{\lambda x : \lambda \neq 0\} \tag{11}$$

Proving π is two-to-one

Given two points $x_1, x_2 \in S^n$, they give the same line if for each point $\lambda_1 y_1 \in \pi(x_1)$ there exists a $\lambda_2 \neq 0$ such that:

$$\lambda_2 x_2 = \lambda_1 x_1 \tag{12}$$

Squaring equation 12 results in

$$\lambda_1^2 x_1^2 = \lambda_2^2 x_2^2 \tag{13}$$

but since $x_1, x_2 \in S^n$, their squares are $x_1^2 = x_2^2 = 1$, and thus

$$\lambda_1 = \pm\lambda_2 \tag{14}$$

Assigning this back into 12 and dividing by λ_1, which is not zero by definition, gives:

$$x_1 = \pm x_2 \tag{15}$$

Meaning there are exactly two points on S^n that gives each line in $\mathbb{R}P^n$, so the map π is a two-to-one map.

Proving π is smooth

The function π^{-1} gives the set of all points responding to a specific line, i.e. the two points shown in the previous subsection $\pm x$. Therefore for a point $x \in S^n$ and an open set $V \subseteq \mathbb{R}P^n$ such that $\pi(x) \in V$, the set $\pi^{-1}(V)$ is an open set made of two neighborhoods around the points $\pm x$.
Now, for a point $x \in S^n$, let (U_i, φ_i) be the chart over S^n that removes the i-th coordinate:

$$\varphi_i : S^n \to \mathbb{R}^n \qquad \varphi_i(x) = (x_1, ..., x_{i-1}, x_{i+1}, ..., x_n) \tag{16}$$

such that $x \in U_i$ and (V_j, ψ_j) a chart in the atlas defined in Question 4 over $\mathbb{R}P^n$ such that $\pi(x) \in V_j$. The composite function $\psi_i \circ \pi \circ \varphi_i^{-1}$ of a point $y \in \mathbb{R}^n$ with coordinates $y_i : i = 1, ..., n$ is:

$$\psi_i \pi \varphi_i^{-1}(y) = \left[1 - \sum_{j \neq i} y_j^2\right]^{-1/2} (y_1, ..., y_{i-1}, y_{i+1}, ..., y_n) \tag{17}$$

Which is smooth for all $y \in \pi^{-1}(V_i) \cap U_i$, since the denominator is non zero. Since $\varphi_i \varphi_j^{-1}$ is smooth for ever i, j as well as $\psi^i \psi_j^{-1}$, we can conclude that $\psi_i \pi \varphi_j^{-1}$ is also smooth for every i, j, which means it holds for all compatible atlases, and the function π is smooth by definition.

proving π is a local diffeomorphism

The inverse of π is locally smooth, since for every point $x \in \mathbb{R}P^n$ and an open set $V \subseteq \mathbb{R}P^n$ such that $x \in V$, $\pi(V)$ gives the set of all lines passing through the origin and through the points in V, which is an open set, and while π^{-1} is not invertible over all of S^n, locally for each point $x \in S^n$ there is a neighborhood V of x such that $\pi^{-1} : \pi(V) \to V$ is one-to-one, making the composite function in equation 17 invertible with a smooth inverse, meaning that π^{-1} is locally smooth, and according to the definition π is locally a diffeomorphism.

Question 6

The structure on $f(N)$ induced by N

Let $f : N \to M$ be an embedding. Then, by definition, it is also a onto its image, and has a bijective inverse f^{-1}. For a chart (U_i, φ_i) on N, define the chart $\left(f(U_i), \varphi_i f^{-1}\right)$ on $f(N)$, where $\varphi_i f^{-1}$ is a composition of bijections and therefore a bijection itself.
For a given atlas on N, the set of all such charts $\left\{\left(f(U_i), \varphi_i f^{-1}\right)\right\}$ matching charts (U_i, φ_i) in the atlas, covers all of $f(N)$, because the atlas covers $\bigcup U_i = N$ and since f is a bijection, $\bigcup f(U_i) = f\left(\bigcup U_i\right) = f(N)$.
In addition, the for each i, j:

$$\varphi_i f^{-1}(f(U_i) \cap f(U_j)) = \varphi_i(U_i \cap U_j) \tag{18}$$

And therefore, by the definition of the atlas on N, $\varphi_i f^{-1}(f(U_i) \cap f(U_j))$ is an open set in $\mathbb{R}^n$.
Finally, for every i, j:

$$(\varphi_i f^{-1}) \circ (\varphi_j f^{-1}) = \varphi_i f^{-1} f \varphi_j^{-1} = \varphi_i \varphi_j^{-1} \tag{19}$$

Which is again, by the definition of an atlas, C^∞. These three points together make $\left\{\left(f(U_i), \varphi_i f^{-1}\right)\right\}$ a smooth atlas on $f(N)$, and the structure that it generates is the smooth structure that makes $f(N)$ a manifold.

The structure on $f(N)$ induced by M

For every point $p \in f(N)$ let (U_i, φ_i) be a chart on M such that $p \in U_i$. Let $\pi : \mathbb{R}^m \to \mathbb{R}^n$ be the projection function, where n and m are the dimensions of N and M respectively. Thus, for an atlas $\{(U_i, \varphi_i)\}$ over M the set of charts $\{(U_i \cap f(N), \pi \circ \varphi_i)\}$ covers $f(N)$, and from the rank theorem, there exist a chart (V_i, ψ_i) for point $x \in N$ such that $\varphi_i \circ f \circ \psi_i^{-1} = \iota$, where ι is the inclusion map from $\mathbb{R}^n$ to $\mathbb{R}^m$ that leaves all other coordinates as zero. Acting on boths sides with the projection π, we get:

$$\begin{aligned} \pi \circ \varphi_i \circ f \circ \psi_i^{-1} &= Id_{\psi_i(V_i)} \\ \pi \circ \varphi_i &= \psi_i \circ f^{-1} \end{aligned} \tag{20}$$

For every two charts (U_i, φ_i) and $\left(U_j, \varphi_j\right)$ of M, the two induced charts on $f(N)$ are $(U_i \cap f(N), \pi \circ \varphi_i)$ and $\left(U_j \cap f(N), \pi \circ \varphi_j\right)$:

$$\begin{aligned} &\pi_{m\to n} \circ \varphi_i \left((U_i \cap f(N)) \cup \left(U_j \cap f(N)\right)\right) = \\ &= \psi_i \circ f^{-1} \left((U_i \cap f(N)) \cup \left(U_j \cap f(N)\right)\right) = \\ &= \psi_i \left(f^{-1}\left(U_i \cup U_j\right) \cap N\right) = \\ &= \psi_i f^{-1}\left(U_i \cup U_j\right) = \\ &= \pi_{m\to n} \circ \varphi_i \left(U_i \cup U_j\right) \end{aligned} \tag{21}$$

Since $\varphi(U_i \cup U_j)$ is an open set, and the projection is surjective and therefore is an open map, $\pi_{m\to n} \circ \varphi_i \left(U_i \cup U_j\right)$ is also an open set. Finally, for each two charts $(U_i \cap f(N), \pi \circ \varphi_i)$ and $\left(U_j \cap f(N), \pi \circ \varphi_j\right)$, their transition map is smooth:

$$(\pi \circ \varphi_i) \circ \left(\pi \circ \varphi_j\right)^{-1} = \pi \circ \varphi_i \circ \varphi_j^{-1} \circ \iota \tag{22}$$

Putting this all together, every smooth atlas on M also induces a smooth atlas on $f(N)$, therefore the structure of M induces a structure on $f(N)$, and $f(N)$ is a submanifold of M.

Proving both structures coincide

Let $\left\{\left(f(U_i), \varphi_i f^{-1}\right)\right\}$ be an atlas induced from N, and let $\left\{\left(V_j, \psi_j\right)\right\}$ be an atlas induced by M. Since f is an embedding, it is also a smooth map, which by definition means $\varphi_i \circ f^{-1} \circ \psi_j|_{f(N)}$ is also smooth. Thus, the two atlases are compatible and their structures coincide.

Question 7

Let $\Delta = \{(p,p) \in M \times M : p \in M\}$ where M is an m-dimensional smooth manifold. From question 1 it is clear that $M \times M$ is a manifold of dimension $2m$. So, to show that Δ is a submanifold of $M \times M$ we must find a smooth structure over Δ for which the inclusion map $\Delta \hookrightarrow M \times M$ is an embedding.
First let us build the atlas for Δ. Let (U_i, φ_i) be a chart of M. Now, let $f : \Delta \to M$ be defined such that $f((p,p)) = p$. It is clear that the set of charts $\left\{\left(f^{-1}(U_i), \varphi_i \circ f\right)\right\}$ is an atlas:

- It covers all of Δ
- For every two charts $\left(f^{-1}(U_i), \varphi_i \circ f\right)$ and $\left(f^{-1}(U_i), \varphi_i \circ f\right)$, the set $\varphi_i \circ f\left(f^{-1}(U_i) \cap f^{-1}(U_j)\right) = \varphi_i\left(U_i \cap U_j\right)$ is an open set.
- For every two charts $\left(f^{-1}(U_i), \varphi_i \circ f\right)$ and $\left(f^{-1}(U_i), \varphi_i \circ f\right)$, their transition map $\varphi_i \circ f \circ \left(\varphi_j \circ f\right)^{-1} = \varphi_i \circ f \circ f^{-1} \circ \varphi_j^{-1} = \varphi_i \circ \varphi_j^{-1}$ is smooth.

This means Δ is a manifold with the structure generated by this atlas. It is also clear from the homeomorphism that it has the same dimension as M.
Next, let us consider the tangent space to Δ. For every point $p \in M$ let T_pM be the tangent space to M at p, and let $v \in T_pM$. For every function $g \in C^\infty(\Delta)$, the differential of f, $df_p : T_pM \to T_{(p,p)\Delta}$ must follow the rule $df_p(v) = v(g \circ f) = v(g, g)$. This gives $df_p(v) = (v, v)$, and the tangent space to Δ at point (p, p) is

$$T_{(p,p)}\Delta = \Big\{(v, v) : v \in T_pM\Big\} \tag{23}$$

Now, showing that the inclusion map $\theta : \Delta \to M \times M$ is an embedding is fairly simple. It is obvious that the inclusion map is smooth, and that it is a homeomorphism onto its image. So, all that is left to show is that it's an immersion. To show this, let us look at its differential $d\theta : T_{(p,p)\Delta} \to T_{(p,p)}(M \times M)$. According to proposition 3.9 in Lee's book, the differential of an inclusion map from a subset of a manifold to the manifold itself is an isomorphism, which is therefore injective. By the definition, θ is an immersion and therefore and embedding, and Δ is thus an embedded submanifold.
The map $\iota : M \to M \times M, \iota(p) = (p, p)$ is then exactly the inverse of f, and as we saw its differential is an isomorphism and therefore injective. It is obviously smooth and a homeomorphism onto its image, which is Δ. This means that ι is an embedding.

Question 8

Let $f : M \to N$ be a smooth immersion that is also injective, and let M be compact. Since N is a manifold, and thus by definition a Hausdorff space, and since an injective function from a compact space to a Hausdorff space is a homeomorphism onto its image, f is a homeomorphism onto its image and therefore an embedding.